

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

Senior engineer with 5+ years of React/TypeScript, Node/NestJS and AWS experience who builds AI features end to end: order-agent pipelines composed on a visual canvas, ranking benchmarked on 15K production aliases, deterministic evals gated in CI, and OpenTelemetry traces. Leads a team of 5 engineers. AI engineering in production since early 2026.

Skills

Frontend: React 18/19, Next.js (SSR), TypeScript, Redux Toolkit, Module Federation, Vite, Nx, Vitest

Backend & Data: Python (FastAPI), Node.js, TypeScript, NestJS, PostgreSQL, Prisma, Hasura (GraphQL), REST APIs, Event-Driven Design

AI / GenAI: LLM Agents, Tool Calling, LangGraph, RAG, Embeddings and Vector Search (OpenSearch kNN, ChromaDB), Prompt Engineering, Structured Outputs (Zod/Pydantic), LiteLLM, Azure OpenAI, LLM Evaluation, Langfuse, OpenTelemetry, OCR / Vision

Cloud & DevOps: AWS (ECS, ECR, Lambda, S3, Amplify), Docker, GitHub Actions, CloudWatch

Engineering: System Design, Idempotency, Multi-Tenant SaaS, Eval-Driven Development, Feature Flags and Roll-back

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Hyderabad. B2B WhatsApp-commerce SaaS; customers include Sun Pharma, UltraTech, ITC, Johnson & Johnson, Veeba

- Led UltraTech's ready-mix concrete (RMC) pour-card and dispatch workflow on WhatsApp end to end with a team of 3, **live in 2.5 weeks**: agent tools, backend service changes and the delivery-challan web app; added an hourly production canary and made dispatch notifications transition-aware so a re-saved challan no longer re-sends
- Redesigned the WhatsApp order agent from one opaque LangGraph block into **seven testable steps** (extract, catalog search, alias lookup, LLM match, quantities, cart, reply) and built the generic “Zo Flow” canvas block, outcome routing and builder validation, so behavior changes per customer need no Python edits or redeploy. Live in production for Veeba, Noida Bakers and field-ops workspaces
- Built the order-agent evaluation system: **6 eval sets, 64 scenarios** graded line by line against ground-truth SKUs (no LLM judge), run twice daily in GitHub Actions with a breach rule (score under 90%, errored item, unfinished run) that alerts Teams and feeds a hosted dashboard
- Replaced a pg_trgm alias lookup that needed every query token to match with a rarity-weighted ranker (exact, initials, typo and prefix matches; IDF over products; coverage gate), benchmarked offline on Veeba's 15,242 production aliases: hit@1 **69% to 78%** leave-one-out and **81% to 95%** on 59 real rep phrasings (misses 6 to 0); shipped behind a flag with an environment rollback
- Built OpenTelemetry-based Langfuse tracing (a span per flow block, traces grouped by conversation, fail-open) and per-step input and output capture in the AI-playground UI, so a failed order turn is diagnosed on one screen instead of in CloudWatch
- Built the knowledge-search tool for the order agent (LiteLLM query embeddings, OpenSearch kNN) and fixed cross-tenant retrieval starvation, where kNN filtered after scoring returned irrelevant chunks, by scoring only the workspace's documents
- Cut catalog-search latency from **about 875 ms to 228 ms per order line** (about 4x) on Veeba production with identical results, by returning a slim index projection
- Promoted to Lead within the first year. Lead a team of 5 engineers and review and merge pull requests for the 19-app monorepo (about 2,000 from 10+ engineers); write the design docs and RCAs that keep work transferable

Earlier: Sumeru Software, Senior Developer (Aug 2022 – Jan 2023); Aerchain, Software Development Engineer 1, converted from intern (Jul 2019 – Jan 2022). Full-stack web apps (React, Node.js) for transaction-based B2B SaaS.

Projects

RoleMiner – job discovery pipeline

Python, FastAPI, ChromaDB, React, Docker

- Scrapes job boards, ranks listings by embedding similarity (TF-IDF fallback), then one LLM call scores the top 20 for fit; a full run costs under \$0.002

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020